

Notes: Deaton & Laroque (RES, 1992)

On the Behaviour of Commodity Prices

Contents

1	Big picture	3
2	Data and measurement (commodity-price facts and key objects)	3
2.1	Why commodity prices are the right object	4
2.2	Commodity price data	4
2.3	Stylized facts in the data	4
2.4	Main state variables in the theory	5
2.5	No-storage benchmark	5
2.6	Empirical moments used to describe prices	6
3	Models and empirical specifications (all equations plus interpretation)	6
3.1	Competitive storage arbitrage condition	6
3.2	Market clearing with inventories	6
3.3	Stationary rational-expectations equilibrium	7
3.4	The cutoff price and the two regimes	7
3.5	Conditional mean restriction	7
3.6	Conditional variance restriction	8
3.7	Renewal-process logic	8
3.8	Comparative statics	8
3.9	Numerical solution and simulation	9
3.10	GMM estimation equation	9
4	Identification and model credibility (what makes the exercise informative)	9
4.1	What identifies the model parameters	10
4.2	Why the authors avoid full-information structural estimation	10
4.3	Monte Carlo check of the estimator	10
4.4	Why the non-negativity constraint is central	11
4.5	What the model explains well	11
4.6	What the model does not fully explain	11
4.7	What would strengthen the evidence	12
5	Tables and figures	12
5.1	Figure 1: Annual sugar prices, 1900–1987	12

5.2	Table 1: Commodity price facts, 1900–1987	13
5.3	Figure 2: Prices with and without storage	13
5.4	Table 2: Distributions of simulated prices	13
5.5	Figures 3 and 4: Price distributions with and without storage	14
5.6	Figure 5: Simulated price series	15
5.7	Table 3: Monte Carlo GMM estimation	15
5.8	Table 4: GMM estimates for actual commodity prices	16
6	Short summary	16
7	Conclusion	17
7.1	Core conclusion	17
7.2	Economic intuition	17
7.3	Takeaway	17

1 Big picture

- **Question.** Can the standard rational-expectations competitive-storage model explain the observed behavior of primary commodity prices?
- **Empirical motivation.** Annual real commodity prices are extremely volatile. They show sharp upward spikes, positive skewness, fat tails, and high short-run autocorrelation, yet they often appear mean-reverting rather than permanently trending.
- **Setting.** The paper studies annual real prices for **13 primary commodities** over **1900–1987**: bananas, cocoa, coffee, copper, cotton, jute, maize, palm oil, rice, sugar, tea, tin, and wheat.
- **Core theory.** The model has competitive risk-neutral inventory holders who can store the commodity from one period to the next. Storage is costly, and inventories cannot be negative.
- **Core nonlinearity.** The aggregate market cannot borrow commodities from the future. This non-negativity constraint on inventories creates two regimes: a storage regime with positive inventories and a stockout regime with zero inventories.
- **Main theoretical contribution.** The paper restates and extends the competitive-storage model, proves existence and uniqueness of a stationary rational-expectations equilibrium, characterizes the two price regimes, derives conditional-mean and conditional-variance restrictions, and gives comparative statics.
- **Main empirical strategy.** Instead of estimating a full structural model by maximum likelihood, the authors first solve and simulate the model numerically, then test a key implication of the model using generalized method of moments.
- **Main success of the model.** Storage can generate positive skewness, occasional violent price spikes, fat tails, and high autocorrelation in normal low-price periods.
- **Main limitation of the model.** The model has difficulty generating the very high autocorrelation observed in many actual commodity-price series. In the simulations, even high-variance specifications produce first-order autocorrelations below the levels seen in many data series.
- **Main empirical result.** For most commodities, the one-period conditional mean restrictions are not strongly rejected. However, the estimated carrying-cost parameters are very close to one, and the model does not fully explain persistence.
- **Economic takeaway.** Competitive storage is a powerful explanation for asymmetric commodity-price behavior, but a stripped-down model with i.i.d. harvest shocks and price data alone is not a complete account of commodity-price dynamics.

2 Data and measurement (commodity-price facts and key objects)

2.1 Why commodity prices are the right object

- Commodity prices matter because many developing countries depend heavily on exports of a few primary commodities.
- Price volatility affects export earnings, fiscal policy, saving, consumption smoothing, agricultural pricing, and risk-sharing arrangements.
- The paper studies whether an economic model, not just a descriptive time-series model, can explain the observed price process.
- The economic issue is not only that commodity prices fluctuate, but that they fluctuate asymmetrically: large upward spikes are more common than equivalent downward crashes.

Interpretation. The paper asks whether a simple arbitrage mechanism—inventory storage—can turn ordinary supply shocks into the unusual price behavior seen in commodity markets.

2.2 Commodity price data

- The data are annual average commodity-price index numbers from the World Bank Commodities Division.
- Prices are deflated by the U.S. consumer price index.
- The sample period is **1900–1987**.
- The paper reports summary facts for **13 commodities**: bananas, cocoa, coffee, copper, cotton, jute, maize, palm oil, rice, sugar, tea, tin, and wheat.
- The main object is the stochastic process for real commodity prices, not quantities or inventories.

Interpretation. The paper intentionally uses price data alone. This makes the exercise demanding, because storage models are easier to test when physical inventory data are also available.

2.3 Stylized facts in the data

- The commodity-price series do not show obvious long-run trends over the century, although the authors do not claim all series are strictly stationary.
- All 13 commodities have first-order autocorrelations above **0.6**; 10 of the 13 are above **0.8**.
- Second-order autocorrelations are lower but still substantial.
- The persistence measure is below **0.3** for all commodities except bananas, suggesting that shocks are not fully permanent.
- Real prices are volatile. Sugar has the highest coefficient of variation, about **0.60**; cocoa is about **0.54**; palm oil is about **0.48**; coffee is about **0.45**; tin is about **0.42**.
- Skewness is non-negative for every commodity in the table. Palm oil has especially high skewness, about **3.24**, and high excess kurtosis, about **16.52**.

- Sugar has a first-order autocorrelation of about **0.62**, a coefficient of variation of about **0.60**, skewness of about **1.49**, and excess kurtosis of about **3.08**.

Interpretation. The empirical target is difficult: the model must explain both smooth persistence in ordinary years and sudden extreme upward price spikes in bad states.

2.4 Main state variables in the theory

- p_t is the commodity price in period t .
- z_t is the random current supply or harvest.
- I_t is the inventory carried from period t into period $t + 1$.
- $D(p_t)$ is deterministic current excess demand.
- $P(z)$ is the inverse demand function, so that without storage the market-clearing price is $p_t = P(z_t)$.
- x_t is the amount on hand:

$$x_t = z_t + (1 - \delta)I_{t-1}. \quad (1)$$

- δ is the physical depreciation or storage-loss rate, so one unit stored today yields $(1 - \delta)$ units tomorrow.
- r is the interest rate and $\beta = 1/(1 + r)$ is the discount factor.

Interpretation. The state variable is not the harvest alone. It is the harvest plus inventories left from the past. This is what allows past shocks to affect current prices.

2.5 No-storage benchmark

- Without inventory holders, market clearing is simply

$$D(p_t) = z_t, \quad (2)$$

or equivalently

$$p_t = P(z_t). \quad (3)$$

- If harvests are i.i.d., prices are also i.i.d. in the no-storage benchmark.
- In that benchmark, price skewness can arise from convex inverse demand, but high autocorrelation cannot arise from i.i.d. harvests.

Interpretation. Storage is the model's source of serial dependence. It carries supply across time and therefore links today's price to yesterday's scarcity or abundance.

2.6 Empirical moments used to describe prices

- **One-year autocorrelation** measures short-run persistence.
- **Two-year autocorrelation** measures whether persistence lasts beyond one period.
- **Persistence** is the normalized spectral density at frequency zero, interpreted as the long-run fraction of a shock that persists.
- **Coefficient of variation** measures unconditional volatility relative to the mean.
- **Skewness** measures asymmetry in the price distribution.
- **Excess kurtosis** measures tail thickness relative to the normal distribution.

Interpretation. The paper does not evaluate the model using a single statistic. It asks whether the model can jointly account for persistence, volatility, asymmetry, and rare extreme realizations.

3 Models and empirical specifications (all equations plus interpretation)

3.1 Competitive storage arbitrage condition

Inventory holders are risk-neutral and can borrow or lend at the interest rate r . Holding I_t units of inventory gives expected profit

$$[\beta(1 - \delta)E_t p_{t+1} - p_t] I_t. \quad (4)$$

Profit maximization implies the complementary-slackness condition

$$I_t = 0 \quad \text{if} \quad \beta(1 - \delta)E_t p_{t+1} < p_t, \quad (5)$$

and

$$I_t \geq 0 \quad \text{only if} \quad \beta(1 - \delta)E_t p_{t+1} = p_t. \quad (6)$$

Interpretation. If the discounted expected future price is below today's price, storage loses money and inventories are zero. If inventories are positive, arbitrage pushes today's price up to the discounted expected future price.

3.2 Market clearing with inventories

Total available supply equals current demand plus inventories carried forward:

$$z_t + (1 - \delta)I_{t-1} - I_t = D(p_t). \quad (7)$$

Equivalently, using the amount on hand $x_t = z_t + (1 - \delta)I_{t-1}$,

$$I_t = x_t - D(p_t) = x_t - P^{-1}(p_t). \quad (8)$$

Combining market clearing and arbitrage gives

$$p_t = \max \{ \beta(1 - \delta)E_t p_{t+1}, P[z_t + (1 - \delta)I_{t-1}] \}. \quad (9)$$

Interpretation. The current price is the larger of two values: the no-storage market-clearing price and the value justified by expected resale through storage. This max operator is the source of the model's nonlinearity.

3.3 Stationary rational-expectations equilibrium

A stationary rational-expectations equilibrium is a price function $f(x)$ such that $p_t = f(x_t)$ and

$$f(x) = \max \{ \beta(1 - \delta)Ef[z + (1 - \delta)(x - P^{-1}\{f(x)\})], P(x) \}. \quad (10)$$

The inventory policy associated with this price function is

$$I(x) = x - P^{-1}\{f(x)\}. \quad (11)$$

Interpretation. Today's price function must be consistent with the future price function that agents expect. This fixed-point problem is why the model has to be solved numerically.

3.4 The cutoff price and the two regimes

The equilibrium has a cutoff price

$$p^* = \beta(1 - \delta)Ef(z). \quad (12)$$

Theorem 1 implies

$$f(x) > P(x) \quad \text{if} \quad P(x) < p^*, \quad (13)$$

and

$$f(x) = P(x) \quad \text{if} \quad P(x) \geq p^*. \quad (14)$$

- **Storage regime.** When the no-storage price is low, the commodity is abundant. Speculators buy and carry inventories forward. The observed price is above the current-use price $P(x)$.
- **Stockout regime.** When the no-storage price is high, the commodity is scarce. Inventories are exhausted, so price equals the current-use price $P(x)$.

Interpretation. The model does not predict a linear autoregression. It predicts a kinked process in which persistence is strong during storage periods and breaks down when inventories hit zero.

3.5 Conditional mean restriction

The model implies two laws of motion. If $p_t \geq p^*$, inventories are zero and

$$p_{t+1} = f(z_{t+1}). \quad (15)$$

If $p_t < p^*$, inventories are positive and

$$p_{t+1} = \frac{p_t}{\beta(1-\delta)} + \eta_{t+1}, \quad E_t \eta_{t+1} = 0. \quad (16)$$

Therefore the conditional expectation is

$$E(p_{t+1} | p_t) = \frac{\min(p^*, p_t)}{\beta(1-\delta)}. \quad (17)$$

Interpretation. Below the cutoff, expected future price rises almost linearly with today's price. Above the cutoff, today's extreme price does not persist mechanically because the market is stocked out and next period depends mainly on the next harvest.

3.6 Conditional variance restriction

When inverse demand $P(x)$ is convex, the model implies

$$\frac{\partial V(p_{t+1} | p_t)}{\partial p_t} \geq 0 \quad (18)$$

up to the stockout cutoff, after which conditional variance is constant.

Interpretation. High current prices signal low inventories. With less inventory available to buffer next period's harvest shock, next period's price becomes more volatile.

3.7 Renewal-process logic

Theorem 2 gives conditions under which the price follows a renewal process.

- If inventories depreciate, very large inventory stocks cannot accumulate forever.
- When the market reaches a stockout, inventories reset to zero.
- After a stockout, the future path starts again from a state governed by the next harvest.

Interpretation. The renewal property is the model's way of combining persistence and mean reversion. Storage transmits shocks across years, but stockouts eventually break the link to the past.

3.8 Comparative statics

Theorem 3 gives several useful comparative statics.

- The equilibrium price function f , the cutoff p^* , and inventory demand $I(x)$ increase with the discount factor β .
- A first-order stochastic increase in supply shocks lowers the equilibrium price function and storage demand.

- If inverse demand is convex, a mean-preserving spread in supply shocks raises prices and inventories.

Interpretation. Storage is more valuable when carrying costs are low and when supply risk is high. Convex demand makes scarcity states especially valuable, increasing the incentive to hold inventories.

3.9 Numerical solution and simulation

The equilibrium price function is computed by iterating on a mapping of the form

$$f_{s+1}(x) = \max \left\{ \beta(1 - \delta) E f_s \left[z + (1 - \delta) (x - P^{-1}\{f_s(x)\}) \right], P(x) \right\}. \quad (19)$$

Given $f(x)$, simulated prices are generated from

$$p_t = f(x_t), \quad x_{t+1} = (1 - \delta) [x_t - P^{-1}\{f(x_t)\}] + z_{t+1}. \quad (20)$$

Interpretation. The model is nonlinear enough that closed-form prices are unavailable. Simulation and numerical invariant distributions are used to learn what the model can generate.

3.10 GMM estimation equation

The empirical test focuses on the conditional-mean implication. Define

$$\gamma = \frac{1}{\beta(1 - \delta)}, \quad \theta = (\gamma, p^*). \quad (21)$$

The innovation is

$$u_t = p_t - \gamma \min(p_{t-1}, p^*). \quad (22)$$

Under the model, u_t is orthogonal to lagged prices. The instrument vector is

$$w_t = (1, p_{t-1}, p_{t-2}, p_{t-3})'. \quad (23)$$

The GMM estimator minimizes

$$u'W(W'W)^{-1}W'u. \quad (24)$$

The overidentifying-restrictions test uses two degrees of freedom because there are four instruments and two estimated parameters.

Interpretation. The GMM exercise asks whether the nonlinear conditional-mean restriction implied by storage is consistent with annual commodity prices.

4 Identification and model credibility (what makes the exercise informative)

4.1 What identifies the model parameters

- The slope parameter $\gamma = 1/[\beta(1 - \delta)]$ is identified from how next year's price responds to this year's price when prices are below the stockout cutoff.
- The cutoff p^* is identified from the point at which the conditional mean stops rising with current price.
- The model's testable content comes from the orthogonality of u_t to lagged prices.
- If the conditional mean is misspecified, lagged prices should help predict the GMM residuals, and the overidentifying restrictions should fail.

Interpretation. The model is not identified from arbitrary serial correlation. It predicts a specific nonlinear forecast function with a cutoff.

4.2 Why the authors avoid full-information structural estimation

- Full-information estimation would require solving the equilibrium price function repeatedly inside the estimator.
- The model has many free components: demand curvature, the harvest distribution, storage costs, and interest costs.
- With price data alone, these components may not be separately identifiable.
- The GMM strategy therefore tests a central implication of the model without requiring full structural recovery of all primitives.

Interpretation. The paper is careful about what price data alone can reveal. It tests whether the model's price dynamics are plausible before trying to estimate every primitive.

4.3 Monte Carlo check of the estimator

- The authors first test the GMM procedure on artificial data generated from the model.
- They draw 500 samples of 100 observations each from a high-variance linear-demand specification.
- The true value of γ is about **1.050**; the mean estimate is about **1.055**; the empirical dispersion is about **0.022**; and the average estimated standard error is about **0.023**.
- The true cutoff is about **114.2**; the mean estimated cutoff is about **112.6**; the empirical dispersion is about **8.79**; and the average estimated standard error is about **13.00**.
- The Durbin–Watson statistic averages about **1.99**, close to the theoretical value of 2.

Interpretation. In data generated by the model, the estimator behaves reasonably well. This makes the empirical failures more informative.

4.4 Why the non-negativity constraint is central

- Without the inventory non-negativity constraint, the storage model would be much closer to a linear smoothing model.
- With the constraint, the market sometimes exhausts inventories and cannot draw commodities from the future.
- In those states, prices can jump sharply because current demand must be rationed by current supply alone.
- The constraint explains why upward spikes are more natural than downward spikes.

Interpretation. The model's asymmetric price behavior is not imposed by asymmetric harvest shocks. It arises from the fact that inventories can be positive but not negative.

4.5 What the model explains well

- It explains why price distributions can be positively skewed even when harvest shocks are symmetric.
- It explains why rare stockouts can generate very large price spikes.
- It explains why prices can be persistent during normal periods with positive inventories.
- It explains why conditional variance should be higher when current prices are high and inventories are low.
- It gives a structural alternative to treating commodity prices as random walks.

Interpretation. The model is successful as a qualitative theory of nonlinear commodity-price dynamics.

4.6 What the model does not fully explain

- Simulated autocorrelations are generally lower than actual autocorrelations in the commodity-price data.
- The GMM estimates of γ are very close to one, implying implausibly small estimates of total carrying costs $r + \delta$.
- Some commodities have too few observed stockout states to identify the cutoff precisely.
- The model assumes i.i.d. harvest shocks, but many commodities may have serially correlated supply shocks.
- The model ignores advance information about harvests, heterogeneous storage technologies, market-specific institutions, government interventions, convenience yields, and risk aversion.

Interpretation. The model captures the asymmetry of prices better than it captures the amount of persistence in prices.

4.7 What would strengthen the evidence

- Direct inventory data would allow sharper tests of the storage mechanism.
- Commodity-specific supply and demand models could explain differences across commodities.
- Allowing serially correlated harvests could help account for high autocorrelation.
- Modeling advance information about future harvests could smooth price responses and alter measured persistence.

Interpretation. The paper is not a final structural estimate of all commodity markets. It is a disciplined test of whether the canonical storage model can explain the main price facts.

5 Tables and figures

5.1 Figure 1: Annual sugar prices, 1900–1987

Read:

- The figure plots the annual real dollar price of sugar from 1900 to 1987.
- The series has no obvious century-long trend.
- It contains several sharp peaks, especially around 1920 and during the late 1970s and early 1980s.
- The 1975 sugar price was nearly ten times its value less than a decade earlier.

Economic message. Commodity prices can spend long periods in relatively normal ranges and then jump violently. A useful theory must explain both ordinary persistence and rare explosions.

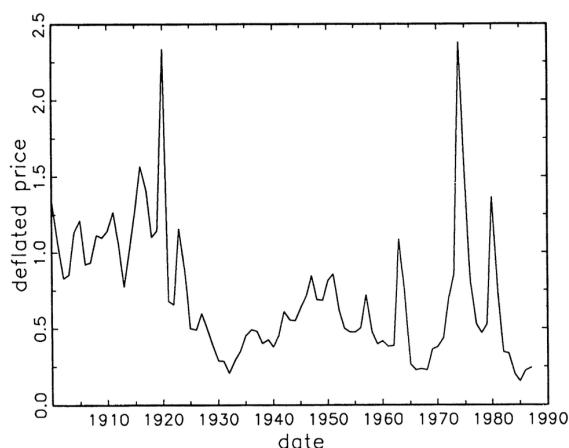


FIGURE 1
Annual sugar prices, 1900–1987

TABLE 1
Commodity price facts 1900–1987

	One-year a-c	Two-year a-c	Persistence	Coeff. of variation	Skewness γ_1	Kurtosis γ_2
bananas	0.91	0.82	0.52	0.17	0.15	-0.98
cocoa	0.83	0.66	0.24	0.54	0.99	1.21
coffee	0.80	0.62	0.11	0.45	1.66	3.82
copper	0.84	0.64	0.22	0.38	1.02	0.86
cotton	0.88	0.68	0.13	0.35	0.35	0.07
jute	0.71	0.45	0.09	0.33	0.61	0.26
maize	0.76	0.53	0.10	0.38	1.18	2.48
palm oil	0.73	0.48	0.05	0.48	3.24	16.52
rice	0.83	0.61	0.08	0.36	0.55	0.03
sugar	0.62	0.39	0.06	0.60	1.49	3.08
tea	0.78	0.59	0.28	0.26	0.04	0.04
tin	0.90	0.76	0.18	0.42	1.66	3.05
wheat	0.86	0.68	0.11	0.38	0.87	0.61

Notes. One-year and two-year a-c are first- and second-order autocorrelation coefficients respectively. Persistence is the normalized spectral density at zero, interpreted as a measure of the degree to which shocks persist; it uses a Bartlett window with a window width of 40 years. The skewness measure γ_1 is $\mu_3/(\mu_2)^{1.5}$ and the kurtosis measure γ_2 is $(\mu_4/\mu_2^2)-3$, where μ_r is the r th (central) moment.

Source. World Bank Commodities Division. The underlying data are index numbers of average prices for each year, deflated by the U.S. consumer price index.

5.2 Table 1: Commodity price facts, 1900–1987

Read:

- All 13 commodities have first-order autocorrelations above 0.6.
- Ten of the 13 commodities have first-order autocorrelations above 0.8.
- Persistence measures are much lower than one for almost all commodities, so a pure random-walk interpretation is not attractive.
- Coefficients of variation are high for many commodities, with sugar around 0.60 and cocoa around 0.54.
- Skewness is never negative and is very high for palm oil, coffee, tin, sugar, maize, cocoa, and copper.
- Palm oil has the most extreme kurtosis, about 16.52.

Economic message. The empirical target is a combination of high autocorrelation, mean reversion, volatility, positive skewness, and fat tails.

5.3 Figure 2: Prices with and without storage

Read:

- The figure compares the no-storage inverse demand price $P(x)$ with the equilibrium storage price $f(x)$.
- When the amount on hand is low, the two prices coincide: inventories are exhausted.
- When the amount on hand is high, the storage price lies above the current-use price because speculators buy and carry inventories forward.
- The gap between $f(x)$ and $P(x)$ is the value of storage.

Economic message. Storage raises prices in abundant states and smooths prices across time, but it cannot prevent spikes when inventories are exhausted.

5.4 Table 2: Distributions of simulated prices

Read:

- The table reports model-implied moments from five simulated specifications.
- Low-variance cases, with no-storage price coefficient of variation near 0.10, generate frequent stockouts around 0.82 and very low first-order autocorrelation around 0.08.
- In the high-variance linear case, the probability of stockout falls to about 0.16, while first-, second-, and third-order autocorrelations rise to about 0.48, 0.31, and 0.22.

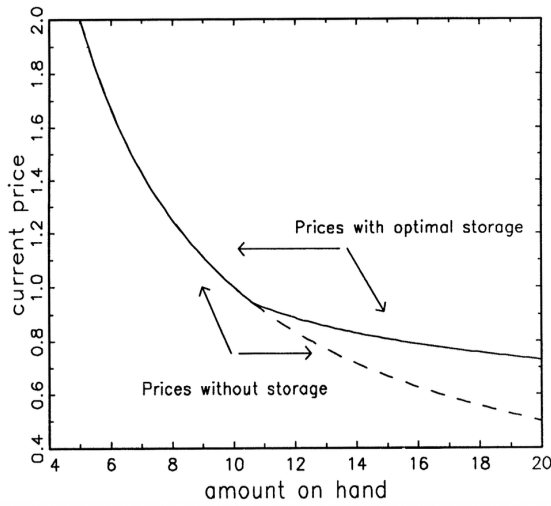


FIGURE 2
Prices with and without storage

Demand	$P(x) = a + bx$ z is $N(100, \sigma = 10)$		$P(x) = x^{-\rho}$ $\ln z$ is $N(0, \sigma^2)$		
Stocks					
Parameters	$r = 0.056$ $\delta = 0.05$ $a = 200$ $b = -1$	$r = 0.05$ $\delta = 0$ $a = 600$ $b = -5$	$r = 0.056$ $\delta = 0.05$ $\rho = 1$ $\sigma = 0.10$	$r = 0.056$ $\delta = 0.05$ $\rho = 1$ $\sigma = 0.50$	$r = 0.056$ $\delta = 0.05$ $\rho = 5$ $\sigma = 0.10$
Coeff of variation ⁽¹⁾	0.10	0.50	0.10	0.53	0.53
Probability of stockouts	0.82	0.16	0.82	0.24	0.24
First-order a.c. ⁽²⁾	0.08	0.48	0.08	0.33	0.29
Second-order a.c. ⁽²⁾	0.01	0.31	0.01	0.16	0.14
Third-order a.c. ⁽²⁾	0.00	0.22	0.00	0.09	0.08
Skewness $\gamma_3^{(3)}$	0.43	1.99	0.61	3.41	3.15
Kurtosis $\gamma_4^{(3)}$	-0.29	5.50	0.17	24.22	16.43
Wilcoxon two-sample pooling test ⁽⁴⁾	0.59	1.27	1.43	-0.05	0.19
	1.44	-3.27	-0.40	0.05	0.29
	2.30	-3.27	-0.76	2.33	-2.57
	-0.16	-1.08	-0.05	-0.20	-0.30
	-1.08	4.37	-0.55	0.33	-1.07

Notes. (1) Coefficient of variation of the price distribution in the absence of inventories. (2) Autocorrelations of prices. (3) $\gamma_1 = \mu_1/(\mu_1)^{1/2}$, $\gamma_2 = (\mu_2/\mu_1^2) - 3$, where μ_r is the r -th central moment. (4) Test statistics as defined, e.g. in Kendall and Stuart (1973), pp. 510-516. To obtain each number, two independent simulations of 2000 periods are calculated for the same model; the (non-parametric) Wilcoxon statistic then tests whether these two realizations come from the same population. The statistic has an $N(0, 1)$ distribution under the null.

- The high-variance linear case also generates skewness around 1.99 and excess kurtosis around 5.50.
- High-variance iso-elastic cases generate stockout probabilities around 0.24, first-order autocorrelations around 0.33 and 0.29, and much higher skewness and kurtosis.

Economic message. The model can generate skewness and fat tails, especially with convex demand. But even specifications with substantial storage have difficulty matching the very high autocorrelations in actual data.

5.5 Figures 3 and 4: Price distributions with and without storage

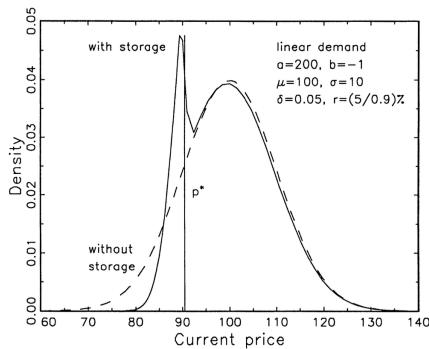


FIGURE 3
Prices with and without storage: linear case

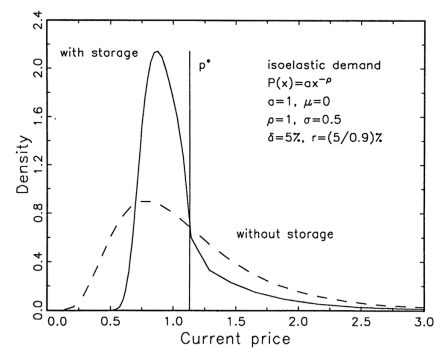


FIGURE 4
Prices with and without storage: iso-elastic case

Read:

- These figures compare the invariant price distribution with storage to the price distribution without storage.

- Storage thins the left tail because speculators buy when prices are low.
- Storage also reduces many high-price outcomes by releasing inventories when supply is scarce.
- Rare stockouts remain possible, and when they occur, the price can be very high.
- With convex iso-elastic demand, the upper tail becomes especially thin but still extends to high prices.

Economic message. Storage makes extreme high prices rare, not impossible. This is why the model can produce occasional spikes within a generally smoothed price process.

5.6 Figure 5: Simulated price series

Read:

- The figure plots a 200-period simulated price series under an iso-elastic demand specification.
- Most observations occur in a low-price, low-volatility range.
- Occasionally the simulated series displays very sharp upward spikes.
- The visual pattern resembles actual price series such as sugar.

Economic message. The competitive-storage model naturally generates “doldrums” punctuated by crises: long periods of buffered prices followed by rare stockouts and large price spikes.

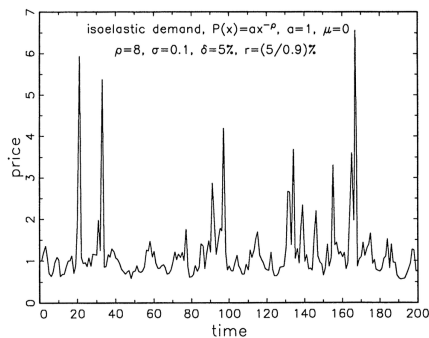


FIGURE 5
200 period simulation of prices

TABLE 3 Monte Carlo GMM estimation of commodity price model				
Parameter or statistic	Theoretical asymptotic value or mean	Average value from 500 replications	Empirical dispersion	Average estimated standard error
γ	1.050	1.055	0.022	0.023
p^*	114.2	112.6	8.79	13.00
OID	2.00	2.64	2.25	—
D.W.	2.00	1.99	0.14	—

5.7 Table 3: Monte Carlo GMM estimation

Read:

- The authors simulate data from the model and estimate the GMM specification on 500 artificial samples.
- The estimated slope parameter is centered close to its true value.
- The estimated cutoff is also centered close to its true value, though its robust standard error is somewhat large.

TABLE 4
GMM estimates of commodity price models

	$\theta^1 = [\beta(1 - \delta)]^{-1}$	S.E.	p^*	S.E.	Stockouts	OID	D.W.
bananas	1.004	0.02	0.694	0.07	0.16	1.17	1.88
cocoa	1.025	0.14	0.345	0.25	0.08	2.31	1.63
coffee	1.040	0.07	0.378	0.11	0.10	2.54	1.45
copper	1.015	0.05	0.792	0.43	0.07	5.35	1.58
cotton	1.013	0.05	0.939	0.30	0.08	10.9	1.21
jute	1.046	0.04	0.763	0.10	0.19	8.17	1.74
maize	1.015	0.08	1.076	0.74	0.06	0.81	1.79
palm oil	0.995	0.14	1.426	2.81	0.01	2.26	1.84
rice	1.012	0.06	0.927	0.34	0.08	6.37	1.43
sugar	1.090	0.14	1.029	0.28	0.23	4.50	1.90
tea	1.009	0.04	0.637	0.15	0.16	0.65	1.77
tin	1.013	0.07	0.449	0.23	0.05	2.71	1.62
wheat	1.005	0.07	1.103	0.47	0.08	5.80	1.36

Notes. The parameters are those given in equation (9). The column labelled stockouts shows the fraction of observations above the estimated cutoff p^* ; the units of p^* and its standard error are those of the underlying price index numbers, and have no direct interpretation. OID is the overidentifying test statistic, and D.W. is the Durbin–Watson test.

- The Durbin–Watson statistic behaves as expected.
- The overidentifying test tends to reject somewhat too often at conventional levels.

Economic message. The estimator is capable of recovering the model’s parameters when the data are generated by the model, but inference about the cutoff can be imprecise when stockouts are rare.

5.8 Table 4: GMM estimates for actual commodity prices

Read:

- The estimated $\gamma = 1/[\beta(1 - \delta)]$ is slightly above one for 12 of the 13 commodities; palm oil is slightly below one.
- None of the slope estimates is significantly different from one.
- The implied total carrying costs are small; the largest point estimate is for sugar, around 9 percent.
- Estimated stockout frequencies are low for many commodities: palm oil is about 0.01, tin about 0.05, maize about 0.06, copper about 0.07, and several others around 0.08–0.10.
- Sugar has the highest estimated stockout frequency, about 0.23.
- The overidentifying test exceeds the 5 percent chi-square critical value for cotton, jute, and rice. Wheat is close to the cutoff.
- Durbin–Watson statistics are low for cotton, rice, wheat, and coffee, suggesting remaining serial dependence for some commodities.

Economic message. The conditional-mean restriction is not massively rejected across the board, but the fitted model still leaves the high autocorrelation of commodity prices insufficiently explained.

6 Short summary

- The paper studies whether the competitive-storage model can explain annual real commodity-price behavior.

- The data show high autocorrelation, substantial volatility, positive skewness, and fat tails.
- The model's key mechanism is the impossibility of negative aggregate inventories.
- This creates a nonlinear two-regime price process: storage when prices are low and stockouts when prices are high.
- The model predicts a kinked conditional mean and rising conditional variance as prices approach the stockout region.
- Simulations show that storage can generate rare price spikes, skewness, and kurtosis.
- GMM estimates test the model's nonlinear conditional-mean implication.
- The model performs reasonably on some moment restrictions but has trouble matching the high autocorrelation in the data.

7 Conclusion

7.1 Core conclusion

The standard rational-expectations competitive-storage model is a strong qualitative theory of commodity-price asymmetry. It explains why prices can be positively skewed, why large upward spikes can occur, and why prices can be persistent in normal times.

7.2 Economic intuition

Storage buffers supply shocks, but the buffer is one-sided. Inventories can be accumulated in abundant periods, but the market cannot hold negative inventories in scarce periods. Once inventories are exhausted, current supply alone must clear the market, producing high and volatile prices.

7.3 Takeaway

The paper's main lesson is that nonlinear storage constraints are essential for understanding commodity prices. But the simplest version of the model, with i.i.d. harvest shocks and price data alone, is not enough to fully explain the observed degree of persistence. A richer model likely needs serially correlated supply, advance information, commodity-specific institutions, or direct inventory data.